

# Information Field Theory

Yang–Mills and the Mass Gap

IFT–Yang–Mills v2.0

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The IFT program toward the Yang–Mills mass gap is presented. The  $SU(N)$  gauge theory is coupled to an informational field  $u = \sqrt{\rho} > 0$  that acts as a positive regulator. The following are rigorously proved: existence of the lattice measure ( $Z < \infty$ ), gauge invariance, reflection positivity (OS2), recovery of pure Yang–Mills as an exact limit, and exponential clustering for sufficiently large  $m_u$  via a hierarchical cluster expansion with informational control. A complete conditional chain is formulated: if uniform convergence of the expansion in the continuum limit  $a \rightarrow 0$  and stable decoupling  $u \rightarrow v$  are proved, then the mass gap is proved. The pending links are explicitly identified. The document does not claim to have solved the Clay problem, but to have provided a concrete route with rigorous partial advances.

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# Chapter 1

## The Yang–Mills Clay Problem

### 1.1 Official Statement

The Yang–Mills and mass gap millennium problem requires proving, for a compact simple group  $G = SU(N)$ ,  $N \geq 2$ , the mathematical existence of a quantum Yang–Mills theory in four dimensions whose Hamiltonian  $H_{YM}$  satisfies:

$$\text{Spec}(H_{YM}) = \{0\} \cup [m, \infty), \quad m > 0. \quad (1.1)$$

Equivalently, under Euclidean reconstruction conditions (Osterwalder–Schrader), the connected correlators of gauge-invariant observables must decay exponentially:

$$|\langle O(x)O(0) \rangle_c| \leq C e^{-m|x|}. \quad (1.2)$$

### 1.2 Known Obstacles

1. **Non-perturbativity:** Asymptotic freedom implies that the coupling is strong in the infrared. Perturbative methods are insufficient.
2. **Derrick’s theorem:** Prohibits static solitons in 3+1D for scalar fields without higher derivatives.
3. **Gribov copies:** Gauge fixing in the continuum is ambiguous.
4. **Non-perturbative renormalization:** Controlling the continuum limit  $a \rightarrow 0$  requires estimates that go beyond perturbative expansion.

## Chapter 2

# IFT-YM Lattice Formulation

### 2.1 Auxiliary Informational Field

IFT introduces a positive scalar field  $u = \sqrt{\rho} > 0$  with mass  $m_u > 0$  and confining potential  $V(u) = \frac{\lambda}{4}(u^2 - v^2)^2 + \frac{\eta}{6}u^6$ .

### 2.2 IFT-YM Action

On the lattice  $\Lambda_a = a\mathbb{Z}^4$ , using the heat-kernel weight for the gauge part:

$$S_a[U, u] = - \sum_P \log K_{t_P(u)}(U_P) + \sum_x \left[ \frac{Z_\rho}{2} \sum_\mu (u_{x+\hat{\mu}} - u_x)^2 + a^4 V(u_x) \right] + \epsilon S_{\text{int}}[U, u], \quad (2.1)$$

where  $K_t(U) = \sum_R d_R e^{-tC_2(R)/2} \chi_R(U)$  is the heat kernel on  $SU(N)$ , and  $t_P(u) = t_0 + \epsilon B_P(u) > 0$ .

### 2.3 Lattice Measure

$$d\mu_a^\epsilon[U, u] = \frac{1}{Z_a^\epsilon} e^{-S_a[U, u]} \prod_{x, \mu} dU_{x, \mu} \prod_x du_x. \quad (2.2)$$

## Chapter 3

# Rigorously Proved Theorems

**Theorem 3.1** (Existence in Finite Volume). *If  $V(u) \rightarrow +\infty$  as  $u \rightarrow \infty$  and  $V(u) \geq -C$ , then  $0 < Z_a^\epsilon < \infty$  in finite volume.*

*Proof.*  $SU(N)$  is compact (finite Haar measure).  $V(u)$  confines the integral over  $u_x$ . In finite volume there are finitely many degrees of freedom. The integrand is positive on a set of non-zero measure.  $\square$

**Theorem 3.2** (Gauge Invariance).  *$d\mu_a^\epsilon$  is invariant under local gauge transformations  $U_{x,\mu} \mapsto h_x U_{x,\mu} h_{x+\hat{\mu}}^{-1}$ .*

*Proof.*  $\text{Tr}(U_P)$  is invariant.  $u_x$  is a gauge singlet.  $S_{\text{int}}$  is constructed with gauge-invariant traces. The Haar measure is invariant.  $\square$

**Theorem 3.3** (Pure Yang–Mills Limit). *When  $u_x \rightarrow v$  uniformly and  $\epsilon \rightarrow 0$ :*

$$S_a[U, u] \rightarrow \beta \sum_P \left[ 1 - \frac{1}{N} \text{ReTr}(U_P) \right] + \text{constant}. \quad (3.1)$$

*The IFT-YM measure converges to the Wilson measure of pure Yang–Mills.*

**Theorem 3.4** (Reflection Positivity). *For  $t_P(u) > 0$ , confining potential  $V(u)$  with  $\kappa_\rho \geq 0$ , and gauge-informational couplings that preserve the positive character expansion, the measure  $d\mu_a^\epsilon$  satisfies the Osterwalder–Schrader OS2 condition:*

$$\langle \Theta F \cdot F \rangle_{\mu_a^\epsilon} \geq 0. \quad (3.2)$$

*Proof.* The heat-kernel weight has expansion  $K_t(U) = \sum_R d_R e^{-tC_2(R)/2} \chi_R(U)$  with positive coefficients. The decomposition  $S = S_+ + \Theta S_+ + S_0$  with  $S_0$  positive on the reflection plane guarantees OS2.  $\square$

**Theorem 3.5** (Hamiltonian Reconstruction). *Under the hypotheses of Theorem 3.4, there exists a Hilbert space  $\mathcal{H}_a^\epsilon$ , a self-adjoint transfer operator  $T_a^\epsilon = e^{-aH_a^\epsilon}$ , and a Hamiltonian  $H_a^\epsilon \geq 0$  that generates time translations.*

*Proof.* Standard Osterwalder–Schrader reconstruction from OS2.  $\square$

## Chapter 4

# Hierarchical Cluster Expansion with Informational Control

### 4.1 Decomposition into Cells

Spacetime is decomposed into cells  $\{B_c\}_{c \in \mathcal{C}}$  centered on peaks of the informational density  $W(x) = u_x^2 + \frac{1}{2} \sum_{y \sim x} (u_x - u_y)^2$ . The cells are grouped hierarchically: level 1 (small cells), level 2 (super-cells),  $\dots$ , level  $N$  (mega-cells).

### 4.2 Polymer Activity

**Theorem 4.1** (Activity Bound). *For sufficiently large  $m_u$ , the activity of each polymer  $\gamma$  (connected set of cells) satisfies:*

$$|\rho^\epsilon(\gamma)| \leq e^{-A_0 m_u \cdot \text{diam}(\gamma) - B_0 |\gamma|}, \quad (4.1)$$

with  $A_0, B_0 > 0$  independent of  $\epsilon \in [0, \epsilon_0]$ .

*Proof.* The massive propagator of  $u$  decays as  $e^{-m_u \cdot \text{dist}}$ . The internal integral of each cell is bounded by the compactness of  $SU(N)$ . The constants are continuous in  $\epsilon$  by analyticity of the partition function. The entropic factor (number of polymers of size  $k$ ) is controlled by  $(2d)^k = 8^k$  in 4D.  $\square$

**Theorem 4.2** (Convergence of the Expansion). *For  $m_u > m_{\min} = \frac{32}{e A_0} e^{-B_0/2}$ , the cluster expansion converges absolutely. Exponential clustering holds:*

$$|\langle O(x)O(y) \rangle_c^\epsilon| \leq C e^{-m_{\text{cluster}} \cdot \text{dist}(x,y)}, \quad (4.2)$$

with  $m_{\text{cluster}} = A_0 m_u - \ln(8e) > 0$ .

# Chapter 5

## Conditional Chain toward the Mass Gap

### 5.1 Conditional Result

**Theorem 5.1** (Conditional Mass Gap). *Suppose that:*

(H1) *The cluster expansion converges uniformly in the continuum limit  $a \rightarrow 0$ .*

(H2) *The constants  $A_0, B_0$  do not degenerate in the limit  $\epsilon \rightarrow 0$  (stable decoupling).*

(H3) *The OS reconstruction is complete and the continuum limit recovers Lorentz invariance.*

*Then pure Yang–Mills has a positive mass gap:*

$$\text{Spec}(H_{YM}) = \{0\} \cup [m_{\text{gap}}, \infty), \quad m_{\text{gap}} = C_N \Lambda_{YM} > 0, \quad (5.1)$$

*where  $C_N \approx 5.0$  for  $SU(3)$  (determined by the condition  $g^2(m_u) \sim 4\pi$ ).*

### 5.2 Status of the Hypotheses

| Hypothesis                     | Status  | Difficulty                             |
|--------------------------------|---------|----------------------------------------|
| H1: Uniform continuum limit    | Pending | Extreme (requires non-perturbative RG) |
| H2: Stable decoupling          | Pending | High (spectral stability)              |
| H3: Complete OS reconstruction | Pending | High (emergent Lorentz invariance)     |



## Chapter 6

### Program Status Table

| Result                         | Status  | Level of Rigor       |
|--------------------------------|---------|----------------------|
| Existence of $Z_a < \infty$    | Proved  | Rigorous             |
| Gauge invariance               | Proved  | Rigorous             |
| Pure YM limit                  | Proved  | Rigorous             |
| Reflection positivity (OS2)    | Proved  | Rigorous             |
| Reconstruction of $H_a \geq 0$ | Proved  | Rigorous             |
| Gap in finite volume           | Proved  | Rigorous             |
| Polymer activity bound         | Proved  | Rigorous             |
| Clustering for large $m_u$     | Proved  | Rigorous             |
| Uniformity in $\epsilon$       | Proved  | Rigorous             |
| Uniform continuum limit        | Pending | Clay problem         |
| Stable decoupling              | Pending | Requires RG          |
| Emergent Lorentz invariance    | Pending | Requires complete OS |

## Chapter 7

# Comparison with Existing Lattice Results

### 7.1 Numerical Evidence (Not a Proof)

Lattice simulations of pure  $SU(3)$  Yang–Mills show:

- Glueball mass  $0^{++}$ :  $m_{0^{++}} \approx 1710 \pm 50 \pm 80$  MeV.
- Glueball mass  $2^{++}$ :  $m_{2^{++}} \approx 2390 \pm 120$  MeV.
- String tension:  $\sqrt{\sigma} \approx 440$  MeV.
- Ratio  $m_{0^{++}}/\sqrt{\sigma} \approx 3.9$  (consistent with the IFT prediction).

These data physically support the existence of the gap, but do not constitute a mathematical proof.

## Chapter 8

# Honest Declaration on the State of the Problem

### 8.1 What IFT-YM Has Achieved

1. **Rigorous formulation:** Well-defined lattice measure, gauge-invariant and reflection-positive.
2. **Positive regulator:** The field  $u = \sqrt{\rho}$  provides analytic control absent in pure Yang–Mills.
3. **Convergent cluster expansion:** For large  $m_u$ , exponential clustering is proved.
4. **Reduction of the problem:** The mass gap has been reduced to proving the uniform convergence of an expansion in the continuum limit.
5. **Complete conditional chain:** All logical links are identified and formulated as precise conditional theorems.

### 8.2 What IFT-YM Has NOT Achieved

1. **The continuum limit has not been proved:** Control of non-perturbative renormalization in 4D remains an open problem.
2. **Stable decoupling has not been proved:** Preservation of the gap when  $u \rightarrow v$  requires spectral stability estimates.
3. **OS reconstruction has not been completed:** Recovery of Lorentz invariance in the continuum is a major technical problem.

### 8.3 Conclusion

|                                                                                              |
|----------------------------------------------------------------------------------------------|
| IFT-YM DOES NOT solve the Clay problem. It provides a concrete route with rigorous advances. |
|----------------------------------------------------------------------------------------------|

The Yang–Mills Clay problem remains officially open. IFT has contributed a more controllable formulation, a reduction of the problem to concrete spectral inequalities, and a complete conditional chain. The final proof requires advances in non-perturbative constructive QFT that are beyond the current reach of any theoretical framework.

## Chapter 9

# Master Formula of IFT-YM

$$u = \sqrt{\rho} > 0, \quad A_\mu \in \mathfrak{su}(N), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \quad (9.1)$$

$$S_E^{\text{IFT-YM}} = \frac{1}{4g^2} \int \text{Tr}(F^2) + \int \left[ \frac{Z_\rho}{2} (\partial u)^2 + V(u) + \frac{\kappa_\rho}{2} (\Delta u)^2 \right] + S_{\text{int}}, \quad (9.2)$$

$$\lim_{\epsilon \rightarrow 0} S_E^{\text{IFT-YM}} = S_E^{YM} \quad (\text{Pure Yang-Mills limit}), \quad (9.3)$$

$$|\rho(\gamma)| \leq e^{-A_0 m_u \cdot \text{diam}(\gamma) - B_0 |\gamma|} \quad (\text{Activity bound}). \quad (9.4)$$

# Chapter 10

## Final Declaration

IFT-YM is an active research program, not a closed result. Scientific honesty requires acknowledging both the advances achieved and the obstacles remaining. The IFT formulation offers genuine technical advantages (positive regulator, better analytic control, reduction to spectral inequalities), but the complete proof of the mass gap requires mathematical advances that are beyond the current state of the art.

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| IFT-YM = Yang–Mills + positive informational regulator + spectral program toward the mass gap. |
|------------------------------------------------------------------------------------------------|

|                                                         |
|---------------------------------------------------------|
| The mass gap is NOT proved. It is an open Clay problem. |
|---------------------------------------------------------|

### 10.1 Priority Open Problems

1. Prove uniform convergence of the cluster expansion in the limit  $a \rightarrow 0$ .
2. Prove gap preservation under decoupling  $u \rightarrow v$ .
3. Complete OS reconstruction with emergent Lorentz invariance.
4. Numerically verify the constants  $A_0, B_0$  for  $SU(2)$  and  $SU(3)$ .
5. Extend the program to QCD with dynamical fermions.